

Python programming with AI tasks no. 1

AI-Powered Customer Service Agent for an Online Store Using Python



AI-Powered Customer Service Agent for an Online Store



1. INTRODUCTION

This project is an AI-powered customer service agent for an online store. The agent uses OpenAI API to understand customer questions, answer FAQs, detect intents, and create support tickets for unresolved issues.

2. FEATURES

- ✓ Use OpenAI API
- ✓ Maintain conversation history
- ✓ Answer FAQs
- ✓ Detect customer intent
 - Billing
 - Technical Support
 - Returns
 - General Inquiry
 - Delivery Issue
- ✓ Create support tickets for unresolved issues
- ✓ Save tickets to a JSON file

3. SYSTEM FLOW

```
graph TD
    A[Customer sends a message] --> B[AI understands the message and detects intent]
    B --> C[AI replies to the customer]
    C --> D[If unresolved, create a support ticket]
    D --> E[Save ticket to tickets.json]
```

4. EXAMPLE INTERACTION

```
Customer Service Agent
Type 'exit' to quit.

Customer: My order hasn't arrived.
Agent: I am sorry to hear that your order has not arrived.
Intent: Delivery Issue
Can you please provide your order number?

Customer: 12345
Agent: Thank you. I have created a support ticket for your missing order.

Ticket ID: CS-1001
Status: Open

A human support agent will review your case soon.

Customer: █
```

5. SUPPORT TICKET (EXAMPLE)

Ticket ID	CS-1001
Issue	Customer order has not arrived
Intent	Delivery Issue
Priority	Medium
Status	Open
Summary	Customer reported that order 12345 has not arrived.

6. TICKETS.JSON (EXAMPLE)

```
[
  {
    "ticket_id": "CS-1001",
    "issue": "Customer order has not arrived",
    "intent": "Delivery Issue",
    "priority": "Medium",
    "status": "Open",
    "summary": "Customer reported that order 12345 has not arrived.",
    "created_at": "2025-05-25 14:30:45"
  },
  {
    "ticket_id": "CS-1002",
    "issue": "Refund for returned product",
    "intent": "Returns",
    "priority": "High",
    "status": "Open",
    "summary": "Customer requested refund for product returned on 2025-05-20.",
    "created_at": "2025-05-25 15:02:10"
  }
]
```

7. TECHNOLOGIES USED

 Python

 OpenAI API

 JSON

 python-dotenv

 Terminal

8. DELIVERABLES

 **Python Source Code**
Complete Python code for the customer service agent.

 **README File**
Setup instructions, features, and how to run the program.

 **Screenshots**
Program running, conversation, tickets.json file.

 **Report (3-5 Pages)**
Description, design, implementation, challenges, conclusion.

1. Introduction

Customer service is an important part of every online business. Customers often need help with orders, payments, returns, refunds, technical problems, and general questions. Traditionally, these requests are handled by human support agents, but artificial intelligence can help **automate** simple customer service tasks.

In this assignment, the goal is to develop an **AI-powered customer service agent using Python** and the OpenAI API (you may choose other AI platforms such as Mistral, Claude, Google Gemini, or others). The agent will communicate with customers, understand their problems, answer frequently asked questions, detect the customer's intent, and create support tickets when the issue cannot be solved immediately.

The project simulates a **real-world online store support system**.

2. Problem Statement

Online stores receive many customer support requests every day. Some questions are simple, such as asking about return policies or payment methods. Other requests require further investigation, such as missing orders, failed payments, or technical problems.

The problem is to design and implement a **Python-based AI customer service agent** that can:

- Understand customer messages
 - Maintain a conversation with the customer
 - Answer common questions
 - Detect the type of customer issue
 - Create support tickets for unresolved problems
 - Save support tickets in a JSON file
-

3. Objectives

The main objectives of this assignment are:

1. To build a customer service chatbot using Python.
 2. To use the OpenAI API for generating intelligent responses.
 3. To maintain conversation history between the customer and the AI agent.
 4. To classify customer requests into different intents.
 5. To answer frequently asked questions.
 6. To create support tickets for unresolved customer problems.
 7. To save ticket information in a JSON file.
 8. To demonstrate how AI can be used in real-world customer service.
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4. System Requirements

The customer service agent must include the following features:

4.1 OpenAI API Integration

The application must use the OpenAI API to generate customer service responses. The API key should be stored securely using an environment variable or .env file.

<https://platform.openai.com/home>

4.2 Conversation History

The system must remember the conversation during a session. This allows the AI agent to respond based on previous customer messages.

Example:

Customer	My order has not arrived.
Agent	I am sorry to hear that. Can you provide your order number?
Customer	12345
Agent	Thank you. I will create a ticket for order 12345

4.3 FAQ Answering

The AI agent should answer common online store questions, such as:

- What is your return policy?
- How long does delivery take?
- What payment methods do you accept?
- How can I track my order?
- How do I contact support?

4.4 Intent Detection

The system should detect the customer's intent. Required intents are:

- Billing
- Technical Support
- Returns
- General Inquiry
- Delivery Issue

Example:

Customer: I was charged twice for my order.

Detected Intent: Billing

Customer: I want to return my product.

Detected Intent: Returns

Customer: The website is not working.

Detected Intent: Technical Support

4.5 Support Ticket Creation

If the issue cannot be solved immediately, the system must create a support ticket.

Each ticket should contain:

- Ticket ID
- Customer issue
- Detected intent
- Priority
- Status
- Conversation summary

Example ticket:

```
{  
  "ticket_id": "CS-1001",  
  "issue": "Customer order has not arrived",  
  "intent": "Delivery Issue",  
  "priority": "Medium",  
  "status": "Open",  
  "summary": "Customer reported that order 12345 has not arrived."  
}
```

4.6 Save Tickets to JSON File

All created support tickets must be saved in a file called:

tickets.json

This file will store all unresolved customer issues for later review by a human support agent.

5. Tools and Technologies

The following tools and technologies will be used:

- Python
 - OpenAI API
 - JSON file handling
 - dotenv for environment variables
 - Visual Studio Code or another Python editor
 - Command line / terminal
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6. Expected Program Flow

1. The program starts and welcomes the customer.
 2. The customer enters a question or problem.
 3. The message is added to the conversation history.
 4. The AI agent detects the customer's intent.
 5. The AI agent answers the customer.
 6. If the issue is unresolved, the system creates a support ticket.
 7. The ticket is saved in tickets.json.
 8. The conversation continues until the customer types exit.
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7. Example Interaction

Customer Service Agent

Type 'exit' to quit.

Customer: My order hasn't arrived.

Agent:

I am sorry to hear that your order has not arrived.

Intent: Delivery Issue

Can you please provide your order number?

Customer: 12345

Agent:

Thank you. I have created a support ticket for your missing order.

Ticket ID: CS-1001

Status: Open

A human support agent will review your case soon.

8. Deliverables

The final submission must include:

8.1 Python Source Code

The project should include the main Python file, for example:

customer_service_agent.py

8.2 README File

The README file should explain:

- Project title
- Project description
- Installation instructions
- How to run the program
- Features of the program
- Example usage

8.3 Screenshots

Screenshots should show:

- The program running in the terminal
- A customer conversation
- A created support ticket
- The tickets.json file

8.4 Report

The report should be 3–5 pages and include:

1. Introduction
 2. Problem statement
 3. System design
 4. Implementation details
 5. Example output
 6. Challenges faced
 7. Conclusion
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9. Evaluation Criteria

The project can be evaluated based on the following criteria:

Criteria	Description
Functionality	The chatbot works correctly
OpenAI API Usage	The program uses the OpenAI API properly

Conversation History	The chatbot remembers the session conversation
Intent Detection	The system correctly identifies customer intent
Ticket Creation	The system creates support tickets
JSON Storage	Tickets are saved correctly in a JSON file
Code Quality	Code is clean and readable
Documentation	README and report are clear
Demonstration	Screenshots show the program working

10. Conclusion

This assignment demonstrates how artificial intelligence can be used to improve customer service in an online store. The AI-powered agent can answer common customer questions, classify customer issues, and create support tickets for unresolved problems. By completing this project, students gain practical experience with Python programming, OpenAI API integration, prompt engineering, intent detection, conversation history, and file handling.

The project is realistic, useful, and suitable for understanding how AI agents can support modern businesses.